

TOPIC 1: PROB & BAYES

1. Basic Probability

Total Prob: $P(A) = \sum_i P(A|B_i)P(B_i)$

Cond Prob: $P(A|B) = \frac{P(A \cap B)}{P(B)}$
Indep: $P(A \cap B) = P(A)P(B) \Leftrightarrow P(A|B) = P(A)$

2. Distributions & Math Core

Binomial (Discrete): $X \sim B(n, p)$

$P(X = r) = C_n^r p^r (1 - p)^{n-r}$

Poisson (Discrete): $X \sim \text{Poi}(\lambda)$

$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$ Total: $\sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^\lambda$

Poisson CDF (Sum to k): $P(X \leq k) = \sum_{x=0}^k \frac{e^{-\lambda} \lambda^x}{x!}$

Poisson-Gamma Integral Link (Shortcut):
(Used to bypass integrating λ in Bayesian posterior)

$\int_0^c \frac{\lambda^n e^{-\lambda}}{n!} d\lambda = 1 - \sum_{x=0}^n \frac{e^{-c} c^x}{x!} = 1 - P(Y \leq n \mid \lambda = c)$

General Form (for λ from 0 to c):

$\int_0^c \lambda^n e^{-\lambda} d\lambda = \frac{n!}{a^{n+1}} \left[1 - \sum_{x=0}^n \frac{(a)^x e^{-a}}{x!} \right]$

Exponential (Cont.): $X \sim \text{Exp}(\lambda)$
 $f(x) = \lambda e^{-\lambda x}$ ($x \geq 0$); $E[X] = \frac{1}{\lambda}$, $\text{Var}(X) = \frac{1}{\lambda^2}$

Gamma / Integration Shortcut:

$\int_0^{\infty} \lambda^n e^{-a\lambda} d\lambda = \frac{n!}{a^{n+1}}$

(λ : rate, n : total count, a : total time)

Int by Parts: $\int u dv = uv - \int v du$
 $\int x^n e^{-ax} dx = -e^{-ax} \sum_{k=0}^n \frac{n! x^{n-k}}{k! a^{k+1}}$ (Compact Form)

Exp CDF (Area): $\int_0^c \lambda e^{-\lambda x} dx = 1 - e^{-\lambda c}$

Exp Tail (Survive): $\int_c^{\infty} \lambda e^{-\lambda x} dx = e^{-\lambda c}$

Linear Shift (u-sub Bypass):

$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b}$, $\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)}$

3. Discrete Bayesian (All PMFs)

Note: All $P(\cdot)$ here are PMFs (Discrete points).

Bayes: $P(S_i|I) = \frac{P(I|S_i)P(S_i)}{\sum_j P(I|S_j)P(S_j)}$

Tabular Method:

1. Prior **PMF**: $P(S_i)$
2. Likelihood **PMF**: $P(I|S_i)$
3. Joint **PMF**: $P(S_i \cap I) = P(I|S_i)P(S_i)$
4. Marginal **PMF**: $P(I) = \sum \text{Joint}$
5. Posterior **PMF**: $P(S_i|I) = \text{Joint}/\text{Marginal}$

4. Binomial Sampling Bayes (All PMFs)

Note: Though f is used, binomial is discrete $\rightarrow$ PMF.

Likelihood **PMF**: $f(r|n, p_i) = C_n^r p_i^r (1 - p_i)^{n-r}$

Posterior **PMF**:

$$P(p_i|r, n) = \frac{p_i^r (1 - p_i)^{n-r} P(p_i)}{\sum_j p_j^r (1 - p_j)^{n-r} P(p_j)}$$

5. Continuous Bayesian (All PDFs)

Note: All $f(\cdot)$ here are PDFs. Use Integration ($\int$) to find probabilities.

Prior **PDF**: $f_\Lambda(\lambda)$

Likelihood **PDF**: $f(x|\lambda)$

Marginal(unconditional) **PDF**: $f_X(x) = \int f(x|\lambda)f_\Lambda(\lambda)d\lambda$

Posterior **PDF**:

$$f_\Lambda(\lambda|x) = \frac{f(x|\lambda)f_\Lambda(\lambda)}{\int f(x|\lambda)f_\Lambda(\lambda)d\lambda}$$

Expected Value: $E[\lambda|x] = \int \lambda \cdot f_\Lambda(\lambda|x)d\lambda$

TOPIC 2: DECISION ANALYSIS & SAMPLING

Decision Analysis: Universal Bayesian Updating

The 4-Step Table Method (Works for ANY evidence/distribution):

1. **Prior** ($P(H_i)$): Initial probability of each state. (Must sum to 1)
2. **Likelihood** ($P(E|H_i)$): Prob. of seeing the new Evidence IF in state H_i . (Calculated via given rules/distributions. Do NOT sum to 1)
3. **Joint: Prior $\times$ Likelihood**. The combined weight.

4. Posterior: Joint/ $\sum$ Joint. The updated probability.

(Normalized to sum to 1)

Crucial: Always use the Posterior probabilities to calculate the NEW EMV for your final decision.

1. Decision Under Uncertainty (No Probs)

Maximax: max(max Payoff) (Optimistic)

Maximin: max(min Payoff) (Pessimistic)

Hurwicz (Realism): $\alpha(\text{Max in Row}) + (1 - \alpha)(\text{Min in Row})$

Minimax Regret: Regret = (Best in Col) - Payoff. Choose act with min(max Regret).

Laplace: Average all payoffs in the row. Choose Max.

2. Prior Analysis (Before Sample)

EMV_{prior}: Expected Monetary Value

$= \sum [P_{\text{prior}}(s_i) \cdot \text{Payoff}(s_i)]$. Choose Max.

EOL_{prior}: Expected Opp Loss $= \sum [P_{\text{prior}}(s_i) \cdot \text{Regret}(s_i)]$.

Choose Min.

Optimal Act a^0 has Max EMV_{prior} (which equals Min EOL_{prior}).

3. Value of Perfect Info (EVPI)

Upper bound for any research cost.

EVWPI (Expected Value With Perfect Info)

$= \sum [P_{\text{prior}}(s_i) \cdot (\text{Best Payoff in } s_i)]$.

EVPI = EVWPI - max(EMV_{prior}).

Shortcut: EVPI is exactly eq to min(EOL_{prior}) of optimal act a^0 !

4. Pipeline: EVSI (Buying Imperfect Info z)

When you buy imperfect info (Sample/Survey z).

Step 1 (Prior): Find optimal act a^0 and its EMV_{prior} without sample.

Step 2 (Sample Prob P(z)): Use Bayes to find total probability of getting survey result P(z).

Step 3 (Posterior EMV): For a specific result z, update state probs to **P_{post}**(**s₁|z**). Calculate new EMVs. Find optimal act a_z^0 and its EMV_{post}(z).

Step 4 (EMV With Sample): Multiply each optimal posterior EMV by the probability of getting that survey result:

$EMVS = \sum [P(z) \cdot \text{EMV}_{\text{post}}(z)]$.

Step 5 (EVSI & ENG\$)

EVSI = EMVS - max(EMV_{prior}).

ENG\$ (Expected Net Gain) = EVSI - Cost of Sample.

Rule: If ENG\$ > 0, buy the sample information!

5. Marginal Analysis (Discrete)

Let P: Prob(Success), k: Cost of under-ordering, h: Cost of over-ordering.

Optimal quantity Q^* satisfies: $P(D < Q^*) = \frac{k}{k+h}$

TOPIC 3: UTILITY THEORY (SINGLE)

1. Risk Attitudes

Let $U(x)$ be the utility function, x be the payoff.

Risk Averse: $U''(x) < 0$ (Concave curve)

Prefers sure amount over lottery. $CE < E[X]$

Risk Neutral: $U''(x) = 0$ (Linear line)

Indifferent. $CE = E[X]$

Risk Seeking: $U''(x) > 0$ (Convex curve)

Prefers lottery over sure amount. $CE > E[X]$

2. Core Metrics

For a lottery L with payoffs x_i and probabilities p_i :

Expected Value (EMV): $E[X] = \sum p_i x_i$

Expected Utility: $E[U(X)] = \sum p_i U(x_i)$

Certainty Equivalent (CE):

The guaranteed return having the same utility as L .

$U(CE) = E[U(X)] \Rightarrow CE = U^{-1}(E[U(X)])$

Risk Premium (π):

The amount "paid" to avoid risk.

$\pi = E[X] - CE$

Risk Averse $\Rightarrow \pi > 0$; Risk Seeking $\Rightarrow \pi < 0$

3. Lottery Pricing (Buying vs Selling)

Let W = Current Wealth (Asset Position), L = Lottery.

Buying Price (B): Max amount willing to pay to acquire the right to play L .

$U(W) = E[U(W - B + L)]$

(Utility of staying with current wealth = Expected utility of paying B and playing L)

Selling Price (S):

Min amount willing to accept to give up the right to play L .

$U(W + S) = E[U(W + L)]$

(Utility of taking cash S and walking away = Expected utility of playing L with current wealth)

4. Common Utility Functions & Price Properties

1. **Exponential Utility**: $U(x) = 1 - e^{-x/R}$ ($R > 0$)

Property: CARA (Constant Absolute Risk Aversion).

Comment: Exhibits the Delta Property. Buying Price = Selling Price ($B = S$). Strictly independent of initial wealth W .

2. **Logarithmic Utility**: $U(x) = \ln(x)$ ($x > 0$)

Property: DARA (Decreasing Absolute Risk Aversion).

Comment: Exhibits the Wealth Effect. As wealth increases, risk aversion decreases. Therefore, Selling Price > Buying Price ($S > B$). Depends on W .

3. Quadratic Utility: $U(x) = ax - bx^2$ ($b > 0, x \leq a/2b$)

Property: IARA (Increasing Absolute Risk Aversion).

Comment: As wealth increases, risk aversion increases (economically counter-intuitive). Therefore, Buying Price > Selling Price ($B > S$). Depends on W .

5. Indifference Lotteries

To find an unknown parameter (e.g., R or α):

If a person is "indifferent" between L_1 and L_2 :

Set $E[U(L_1)] = E[U(L_2)]$ and solve for the unknown.

TOPIC 4: MULTI-ATTRIBUTE UTILITY

1. Independence Conditions

Let $u(x, y)$ be utility of attribute X and Y .

Additive Independence (AI):

$u(x, y) = k_x u_x(x) + k_y u_y(y)$ where $k_x + k_y = 1$.

Condition: Preferences over lotteries on X, Y depend only on marginal distributions.

Utility Independence (UI):

X is UI of Y if preferences over lotteries on X do not depend on fixed y .

Mutual Utility Independence (MUI):

X is UI of Y AND Y is UI of X .

2. Multiplicative Form (MUI)

If X, Y are MUI, total utility is:

$u(x, y) = k_x u_x(x) + k_y u_y(y) + k_{xy} u_y(y) + k_{ux} u_x(x)$

where $k = 1 - k_x - k_y$.

Scaling: $u(x^*, y^*) = 1, u(x^0, y^0) = 0$.

$u_i(x_i^*) = 1, u_i(x_i^0) = 0$.

Finding k_i : $k_x = u(x^*, y^0), k_y = u(x^0, y^*)$.

3. Evaluation Steps

Step 1: Define $u_x(x)$ and $u_y(y)$ based on Topic 3 (Risk Attitude).

Step 2: Solve for k_x, k_y using boundary conditions or indifference points.

Step 3: Determine $k = 1 - k_x - k_y$.

Step 4: Calculate $u(x, y)$ for all choices and rank by $E[u(x, y)]$.

TOPIC 5: ANALYTIC HIERARCHY (AHP)

1. Minimize CI & Unknowns (KFC Yiu's Trick)

1. **The 3x3 Transitive Rule (Primary)**:

To minimize CI ($CI \rightarrow 0$), force perfect consistency:

a_{1k} = a_{1j} $\times$ a_{j_k} (e.g., $a_{13} = a_{12} \times a_{23}$, solve for x, y)

Proof: By AM-GM inequality, $3 \times 3 \lambda_{\text{max}}$ is minimized when sum of reciprocal cycles $k + 1/k \geq 2$ (where $k = \frac{a_{12} \times a_{23}}{a_{13}}$) reaches minimum at $k = 1$.

2. **The 1-9 Scale Boundary (Edge Case)**: Saaty scale is strictly 1 to 9. If Transitive Rule yields $a_{ij} > 9$ (e.g., $4 \times 3 = 12$), you MUST cap it: **a_{ij} = 9** (closest valid num).

3. The 4x4 Multi-Path Conflict:

If reaching node D from A yields conflicting path results (e.g., via $B \rightarrow 6$, via $C \rightarrow 8$), use **Geometric Mean**:

$a_{*14} = \sqrt{\text{Path 1} \times \text{Path 2}} = \sqrt{6 \times 8} \approx 6.93$.

2. Standard 6-Step Calculation (e.g., 3x3)

Given Matrix A .

Step 1: Col Sum (C_j)

$C_1 = a_{11} + a_{21} + a_{31}, C_2 = \dots, C_3 = \dots$

Step 2: Normalize (a'_{ij})

Divide each element by its Col Sum: $a'_{ij} = a_{ij}/C_j$

Step 3: Weights Vector (w_i)

Row average of Normalized Matrix:

$w_1 = (a'_{11} + a'_{12} + a'_{13})/3$ (Same for w_2, w_3)

Step 4: Weighted Sum Vector (v_i)

Multiply original A by w :

$v_1 = a_{11}w_1 + a_{12}w_2 + a_{13}w_3$ (Same for v_2, v_3)

Step 5: Max Eigenvalue (λ_{max})

$\lambda_{\text{max}} = \frac{1}{3}(v_1/w_1 + v_2/w_2 + v_3/w_3)$

Step 6: Test CI and CR

$CI = \frac{\lambda_{\text{max}} - n}{n - 1}$ ($3 \times 3 \rightarrow n = 3$)

$CR = CI/RI$ (Accept if < 0.10 . $RI_{n=3} = 0.58, RI_{n=4} = 0.90$)

3. Reverse Engineering (Weights $\rightarrow$ Matrix)

If w_1, w_2, w_3 are given and Matrix A is unknown:

Assume perfectly consistent: $a_{ij} = w_i/w_j$.

(e.g., $a_{12} = w_1/w_2, a_{31} = w_3/w_1$)

TOPIC 6: GAME THEORY (PIPELINE)

Two-Person Zero-Sum Game

• Payoff matrix $A = [a_{ij}]$: amount Player I wins from Player II.

• $\sum$ payoffs = 0 for all outcomes.

Gain Floor & Loss Ceiling (exist for ALL games)

Gain Floor (maximin): $v_I' = \max_i \left\{ \min_j a_{ij} \right\}$

Loss Ceiling (minimax): $v_{II}' = \min_j \left\{ \max_i a_{ij} \right\}$

Expected Gain Floor & Expected Loss Ceiling (Mixed Strategies)

Expected Gain Floor: $v_I = \max_x \left\{ \min_y x A_{\cdot y} \right\}$

Expected Loss Ceiling: $v_{II} = \min_y \left\{ \max_x A_{i \cdot} y^T \right\}$

Minimax Inequality: $v_I' \leq v_I = v_{II} \leq v_{II}'$

If $v_I' = v_{II}' = v$, then a **saddle point** exists.

Saddle Point: An entry a_{ij} that is the **minimum in its row** and **maximum in its column**.

Minimax Theorem: Every finite two-person zero-sum game has optimal mixed strategies, and

$\max_x \min_y x A y^T = \min_y \max_x x A y^T = v$ where v is the **value of the game**.

Dominance:

- Row k dominates Row l if $a_{kj} \geq a_{lj}$ for all j .
- Column k dominates Column l if $a_{ik} \leq a_{il}$ for all i .

Step 1: Find Saddle Point (Pure)

1. **Row (Maximin)**: Find min of each row $\rightarrow$ Pick the MAX.

2. **Col (Minimax)**: Find max of each col $\rightarrow$ Pick the MIN.

If Maximin = Minimax = v , stop! Optimal is pure (1s and 0s).

Step 2: Dominance (Reduce Matrix)

Row wants MAX: If Row $A \geq$ Row $B \Rightarrow$ **DELETE**

SMALLER Row B .

Col wants MIN: If Col $A \leq$ Col $B \Rightarrow$ **DELETE LARGER**

Col B .

Repeat until 2x2. Remember which rows/cols you deleted!

Step 3: Solve 2x2 (Matrix Method & Shortcut)

Matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Let $J = \begin{bmatrix} 1 & 1 \end{bmatrix}$, $A^* = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$,

$|A| = ad - bc$.

Formulas: $x = \frac{JA^*}{JA^*JT}, y = \frac{J(A^*)^T}{JA^*JT}, v = \frac{|A|}{JA^*JT}$

(Shortcut: Denom $\Delta = a + d - b - c$.

Row $x \Rightarrow \text{sum cols}/\Delta$. Col <

23/24 Q1: Decision Criteria & Decision Trees
[Q1a] Returns depend on investment size and market state.

Size	Good	Fair	Poor
S	5k	2k	-1k
M	8k	3k	-2k
L	11k	4k	-3k
VL	15k	1k	-10k

All states equally likely.
[Sol 1a] *Prob:* $P(G) = P(F) = P(P) = 1/3$.

EMV: $EMV(S) = \frac{5k+2k-1k}{3} = 2000$,

$EMV(M) = \frac{8k+3k-2k}{3} = 3000$,

$EMV(L) = \frac{11k+4k-3k}{3} = 4000$,

$EMV(VL) = \frac{15k+1k-10k}{3} = 2000$.

$\Rightarrow$ choose **Large**.

EOL: best-by-column = $(15k, 4k, -1k)$.

Size	G regret	F regret	P regret
S	10k	2k	0
M	7k	1k	1k
L	4k	0	2k
VL	0	3k	9k

$EOL(S) = 4000$, $EOL(M) = 3000$, $EOL(L) = 2000$,
 $EOL(VL) = 4000$.
 $\Rightarrow$ choose **Large**.

EVPI: $EPPI = \frac{15k+4k-1k}{3} = 6000$,

$EVPI = 6000 - 4000 = 2000$.

Rule: $EVPI = \min EOL$.

[Q1b] New product: $F(0.6) \rightarrow +2000$, $U(0.4) \rightarrow -1500$; forecast

cost = 200;

$P(f|F) = 0.8$, $P(u|U) = 0.7$.

[Sol 1b] **No info:** $EMV(M) = 0.6(2000) + 0.4(-1500) = 600$,

$EMV(N) = 0$. So baseline = 600.

Bayes: $P(f) = 0.8(0.6) + 0.3(0.4) = 0.60$, $P(u) = 0.40$.

$P(F|f) = 0.80$, $P(U|f) = 0.20$; $P(F|u) = 0.30$, $P(U|u) = 0.70$.

(i) *With info, gross:*

f : $EMV(M) = 0.8(2k) + 0.2(-1.5k) = 1300$, $EMV(N) = 0$

$\Rightarrow M$.

u : $EMV(M) = 0.3(2k) + 0.7(-1.5k) = -450$, $EMV(N) = 0$

$\Rightarrow N$.

$EMVS = 0.6(1300) + 0.4(0) = 780$, net = $780 - 200 = 580 < 600$.

Optimal: do not buy forecast; choose Market.

(ii) $EVSI = 780 - 600 = 180$,

$EPPI = 0.6(2000) + 0.4(0) = 1200$, $EVPI = 1200 - 600 = 600$.

(iii) *Refund if wrong:*

f : $EMV(M) = 0.8(2k - 200) + 0.2(-1.5k) = 1140$,

$EMV(N) = 0.8(-200) = -160 \Rightarrow M$.

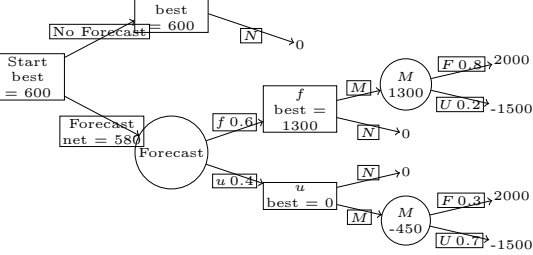
u : $EMV(M) = 0.3(2k) + 0.7(-1.5k - 200) = -590$,

$EMV(N) = 0.7(-200) = -140 \Rightarrow N$.

Net = $0.6(1140) + 0.4(-140) = 628 > 600$.

Optimal: buy forecast; if f then Market, if u then No Market.

Q1b Tree:



M =Market, N =No Market. Root compares net values: No Forecast 600 vs Forecast 580.

Q2: Continuous Distributions & Utility

[Q2a] $G(0.6)$: $f(x) = 9xe^{-3x}$; $B(0.4)$: $f(x) = \frac{1}{2}e^{-x/2}$. Find

(i) $P(X > 2)$, (ii) $P(G|X < 2)$.

[Sol 2a] (i) $P(X > 2|G) = \int_2^\infty 9xe^{-3x} dx = 7e^{-6}$.

$P(X > 2|B) = \int_2^\infty \frac{1}{2}e^{-x/2} dx = e^{-1}$.

$P(X > 2) = 0.6(7e^{-6}) + 0.4(e^{-1}) = 4.2e^{-6} + 0.4e^{-1} \approx 0.147$.

(ii) $P(X < 2) = 1 - P(X > 2) \approx 0.853$, $P(X < 2|G) = 1 - 7e^{-6}$.

$P(G|X < 2) = \frac{0.6(1 - 7e^{-6})}{1 - 4.2e^{-6} - 0.4e^{-1}} \approx 0.700$.

[Q2b] $u(x_1, x_2) = 0.5u_1 + 0.7u_2 + k_3u_1u_2$. Determine dependence, find k_3 , then rank: $L_1 = (30, 100)$, $L_2 = 0.5(30, 100) + 0.5(0, 0)$, $L_3 = 0.6(30, 100) + 0.4(0, 0)$, $L_4 = (30, 0)$, $L_5 = (0, 100)$, $L_6 = 0.4(30, 0) + 0.6(0, 100)$.

[Sol 2b] Dependent because $k_3 \neq 0$.

Set $u(30, 100) = 1$, $u(0, 0) = 0$: $1 = 0.5 + 0.7 + k_3 \Rightarrow k_3 = -0.2$.
 $u(L_1) = 1$, $u(L_2) = 0.5$, $u(L_3) = 0.6$, $u(L_4) = 0.5$, $u(L_5) = 0.7$,
 $u(L_6) = 0.4(0.5) + 0.6(0.7) = 0.62$.

Rank: $L_1 \succ L_5 \succ L_6 \succ L_3 \succ L_2 \sim L_4$.

Q3: AHP & Risk Attitude

[Q3a] $A = \begin{pmatrix} 1 & 2 & 6 \\ a & c & d \\ b & 1/3 & 1 \end{pmatrix}$, $A_{norm} = \begin{pmatrix} 0.6 & 0.6 & 0.4615 \\ 0.3 & 0.3 & 0.4615 \\ 0.1 & 0.1 & 0.0769 \end{pmatrix}$.

Find a, b, c, d ; then CI, CR; then fix for consistency.

[Sol 3a] (i) Col 1 sum = $1/0.6 = 5/3$:

$1 + a + b = 5/3 \Rightarrow a = 0.5$, $b = 1/6$.

Col 2 sum = $2/0.6 = 10/3$: $2 + c + 1/3 = 10/3 \Rightarrow c = 1$.

Col 3 sum $\approx 6/0.4615 = 13$: $6 + d + 1 = 13 \Rightarrow d = 6$.

(ii) Row-average weights: $W = (0.5538, 0.3538, 0.0923)^T$.

$AW = (1.8152, 1.1845, 0.3026)^T$,

$\lambda_{max} = \frac{1}{3} \left(\frac{1.8152}{0.5538} + \frac{1.1845}{0.3538} + \frac{0.3026}{0.0923} \right) = 3.301$.

$CI = \frac{3.301-3}{2} = 0.1505$, $CR = \frac{0.1505}{0.58} = 0.2595 > 0.1$

(inconsistent).

(iii) Perfect consistency needs $a_{12}a_{23} = a_{13}$, so $2d = 6 \Rightarrow d = 3$.

[Q3b] $u(x) = \ln x$, wealth = 30000.

L_1 : probability α of losing 20000, else 0; L_2 : certain loss 3000.

[Sol 3b] Risk attitude: $u''(x) = -1/x^2 < 0 \Rightarrow$ **risk-averse**.

Indifference $L_1 \sim L_2$:

$$\alpha \ln(10k) + (1 - \alpha) \ln(30k) = \ln(27k)$$

$$\alpha \ln(1/3) = \ln(0.9) \Rightarrow \alpha = \frac{\ln(0.9)}{\ln(1/3)} \approx 0.0959$$

$E(W_{L1}) = 0.0959(10k) + 0.9041(30k) = 28082$, $CE = 27000$.

Risk premium = $28082 - 27000 = 1082$.

Q4: Expected Opportunity Loss & Game Theory

[Q4a] Defect rates: A(1%, 0.6), B(3%, 0.3), C(5%, 0.1).

Event	Reject a_1	Accept a_2
A	-\$1500	\$2000
B	-\$500	\$100
C	\$1500	-\$2500

Find (i) EVPI, (ii) verify $\bar{v}_t + \bar{I}_t = c$, (iii) after sampling 10 with 2 defectives, accept or reject?

[Sol 4a] (i)

$EMV(a_1) = 0.6(-1500) + 0.3(-500) + 0.1(1500) = -900$.

$EMV(a_2) = 0.6(2k) + 0.3(100) + 0.1(-2.5k) = 980$.

$EPPI = 0.6(2000) + 0.3(100) + 0.1(1500) = 1380$,

$EVPI = 1380 - 980 = 400$.

(ii) Regrets using best-by-state $(2000, 100, 1500)$:

a_1 : $(3500, 600, 0)$, a_2 : $(0, 0, 4000)$.

$\bar{I}_t(a_1) = 0.6(3500) + 0.3(600) = 2280$, $\bar{I}_t(a_2) = 0.1(4000) = 400$.

$\bar{v}_t(a_1) + \bar{I}_t(a_1) = -900 + 2280 = 1380$,

$\bar{v}_t(a_2) + \bar{I}_t(a_2) = 980 + 400 = 1380$.

(iii) Bayesian update with likelihood $L \propto p^2(1 - p)^8$:

$J(A) = 0.6(0.000092) = 0.000055$,

$J(B) = 0.3(0.000705) = 0.000211$,

$J(C) = 0.1(0.001658) = 0.000166$.

$\sum J = 0.000432$. Hence $P(A|r = 2) = 55/432 \approx 0.128$,

$P(B|r = 2) = 211/432 \approx 0.489$, $P(C|r = 2) = 166/432 \approx 0.383$.

$EMV(a_1) = 0.128(-1.5k) + 0.489(-500) + 0.383(1.5k) = 138$.

$EMV(a_2) = 0.128(2k) + 0.489(100) + 0.383(-2.5k) = -653$.

Decision: Reject a_1 .

[Q4b] $A = \begin{pmatrix} 4 & 8 & 2 & 10 \\ 2 & 4 & 5 & 7 \\ 2 & 8 & 5 & 6 \\ 2 & 1 & 1 & 7 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 8 & 3 & 5 \\ 2 & 6 & 2 & 6 \\ 4 & 1 & 2 & 9 \\ 3 & 3 & 2 & 5 \end{pmatrix}$. Find

saddle point, optimal strategies, and game value.

[Sol 4b] **Matrix A:** row mins = $\{2, 2, 2, 1\}$ so gain floor = 2; col

maxes = $\{4, 8, 5, 10\}$ so loss ceiling = 4. No saddle point.

Domination: delete R4, R2, C4, C2. Reduced game: $\begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$.

Let Player I use $(p, 1 - p)$:

$4p + 2(1 - p) = 2p + 5(1 - p) \Rightarrow p = 3/5$.

Value = $\frac{4 \cdot 5 - 2 \cdot 2}{4 + 5 - 2 - 2} = 16/5 = 3.2$.

Optimal strategies: Player I = $(3/5, 0, 2/5, 0)$, Player II =

$(3/5, 0, 2/5, 0)$.

Matrix B: row mins = $\{3, 2, 1, 2\}$, col maxes = $\{5, 8, 3, 9\}$.

Gain floor = loss ceiling = 3, so saddle point at $(1, 3)$ and game

value = 3.

22/23 Q1: Decision Criteria & Decision Trees

[Q1a] Returns depend on size and market. $y > 0$, states equally likely.

Size	Good	Fair	Poor
S	400	300	200
M	3y	2y	y
L	900	800	100

(i) Range of y for Maximax $\rightarrow$ Large. (ii) Maximin $\rightarrow$ Medium.

(iii) EMV decision. (iv) Minimax regret decision.

[Sol 1a] (i) **Maximax:** Max payoffs are $S=400$, $M=3y$, $L=900$. For

L to be optimal: $900 > 3y \Rightarrow 0 < y < 300$.

(ii) **Maximin:** Min payoffs are $S=200$, $M=y$, $L=100$. For M to be

optimal: $y > 200$ and $y > 100 \Rightarrow y > 200$.

(iii) **EMV** ($200 < y < 300$): $P = 1/3$. $EMV(S) = 300$,

$EMV(M) = 2y \in (400, 600)$, $EMV(L) = 600 \Rightarrow$ **Large**.

(iv) **Regret** ($200 < y < 300$): Max in states $\rightarrow G(900)$, $F(800)$,

$P(y)$. Regret S : $\max(500, 500, y-200) = 500$. Regret M :

$\max(900-3y, 800-2y, 0)$. Since $y > 200$,

$800-2y > 900-3y \Rightarrow 800-2y$ (range 200-400). Regret L :

$\max(0, 0, y-100) = y-100$ (range 100-200). Minimax regret =

$\min(500, 800-2y, y-100) = y-100 \Rightarrow$ **Large**.

[Q1b] Hot drinks: $W(-2000)$, $C(10000)$. Ice creams: $W(8000)$,

$C(-1000)$. Prior: 50% C , 50% W . Forecast cost \$500.

$P(c|C) = 0.9$, $P(w|W) = 0.8$. Maximize profits, find EVSI &

EVPI.

[Sol 1b] **No info:** $EMV(Hot) = 0.5(10k - 2k) = 4000$.

$EMV(Ice) = 0.5(8k - 1k) = 3500$. Base = 4000 (Hot).

Bayes: $P(c) = 0.9(0.5) + 0.2(0.5) = 0.55$,

$P(w) = 0.1(0.5) + 0.8(0.5) = 0.45$.

$P(C|c) = 9/11$, $P(W|c) = 2/11$; $P(C|w) = 1/9$, $P(W|w) = 8/9$.

With info, gross:

c : $EMV(H) = \frac{9}{11}(10k) + \frac{2}{11}(-2k) = 7818.2 \Rightarrow H$.

w : $EMV(I) = \frac{1}{9}(-1k) + \frac{8}{9}(8k) = 7000 \Rightarrow I$.

Gross EMV = $0.55(8600/11) + 0.45(7000) = 4300 + 3150 = 7450$.

Net = $7450 - 500 = 6950 > 4000$.

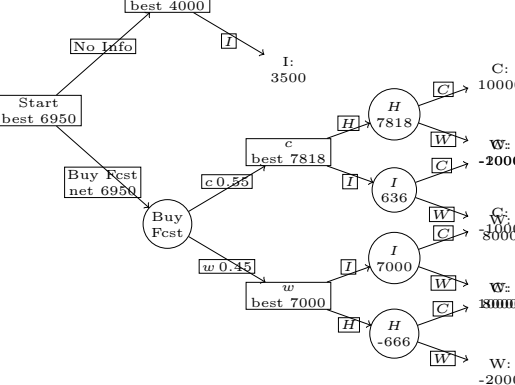
Optimal: Buy forecast; if c sell Hot, if w sell Ice.

$EVSI = 7450 - 4000 = 3450$.

$EPPI = 0.5(10k) + 0.5(8k) = 9000$.

$EVPI = 9000 - 4000 = 5000$.

Q1b Tree:



Q2: Continuous Distributions & Multi-attribute Utility

[Q2a] $f(x|\lambda) = \lambda e^{-\lambda x}$, $f_A(\lambda) = \frac{\lambda e^{-\lambda/2}}{4}$ ($\lambda \geq 0$). (i)

Unconditional pdf, $P(X < 1)$? (ii) If $X = 1.5$, $P(X < 2)$?

[Sol 2a] (i)

$f(x) = \int_0^\infty \lambda e^{-\lambda x} \frac{\lambda e^{-\lambda/2}}{4} d\lambda = \frac{1}{4} \int_0^\infty \lambda^2 e^{-\lambda(x+0.5)} d\lambda$.

Shortcut: $\int_0^\infty \lambda^n e^{-a\lambda} d\lambda = \frac{n!}{a^{n+1}}$. Here $n = 2$, $a = x + 0.5$.

$f(x) = \frac{1}{4} \frac{2!}{(x+0.5)^3} = \frac{1}{(2x+1)^3}$. $P(X < 1) =$